

Abstract

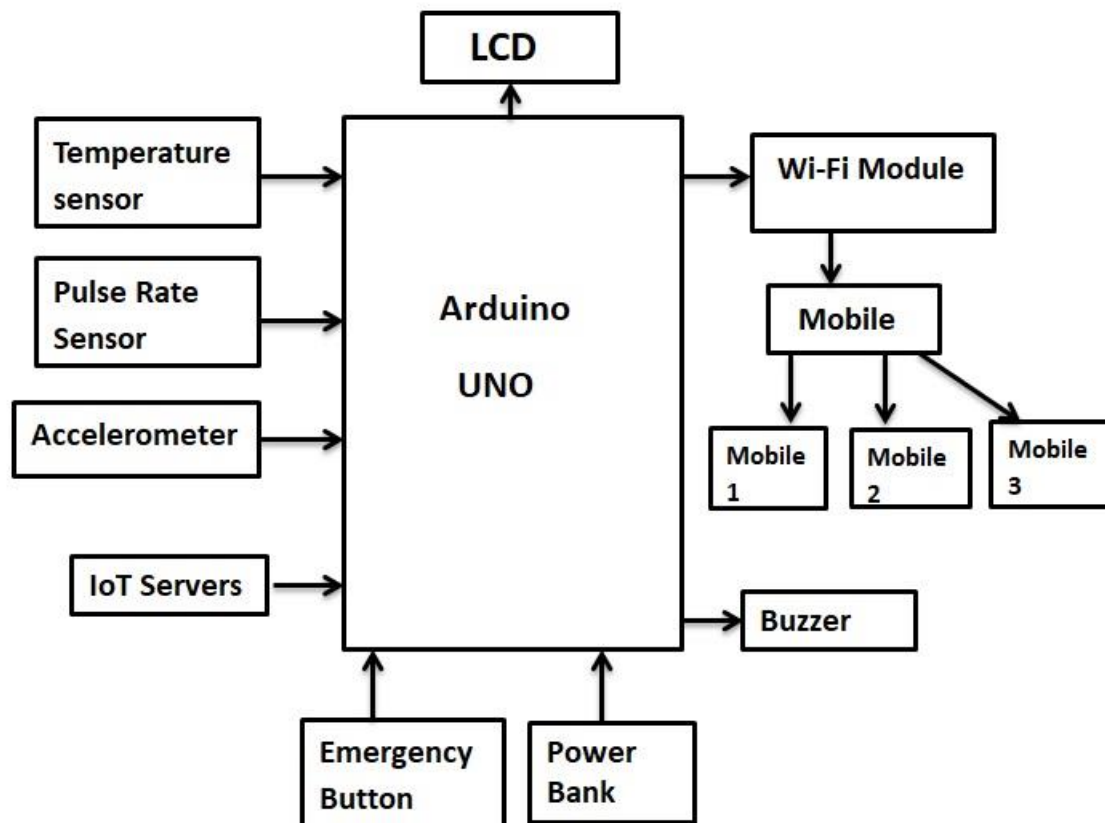
Project Title – IOT based- Life Health Band -Health Monitoring System for Elderly People.

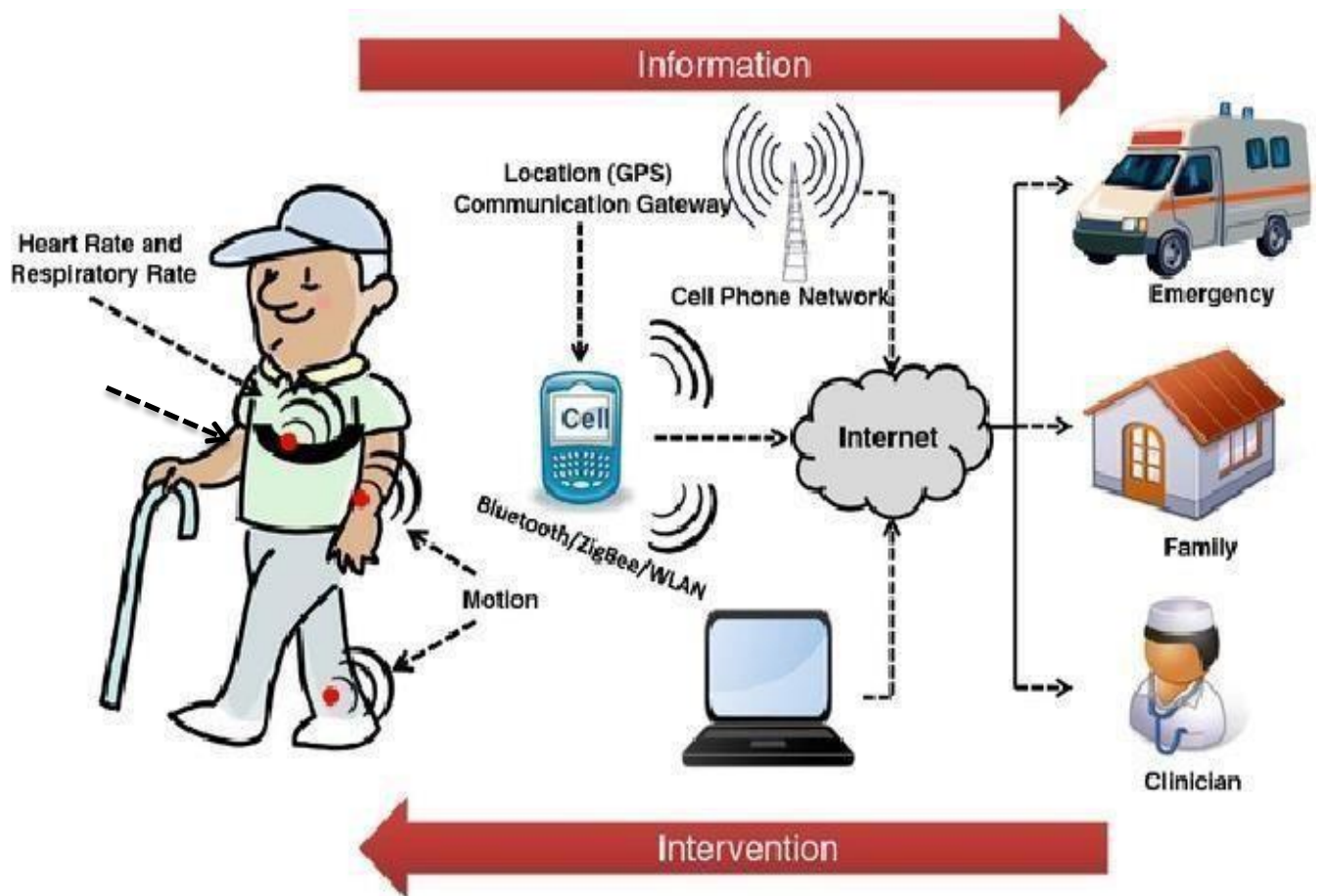
Type – In- House

Introduction –

Old aged people, Alzheimer's affected ones, persons with heart problems always need an attention. If there is a situation that no one is present at home, during these times no one will be there to take care of them. This may lead to situations that we lose them. Health Band is an innovative solution for detecting a person whose health in grave danger. It's mostly designed as a remote rescue system for people who are at risk of having stroke, cardiac arrest and heart attack. Health Band is a mobile health monitoring bracelet capable of reading a human's vital signs. The Internet of Things (IoT) has been widely used to interconnect the available medical resources and offer smart, reliable, and effective healthcare service to the elderly people. Health monitoring for active and assisted living is one of the paradigms that can use the IoT advantages to improve the elderly lifestyle.

Block Diagram –





Working –

We propose a portable and customizable IoT system that can be used to collect the data needed to facilitate the independent living of senior and challenged citizens to improve their quality of life. In addition to the above medical sensor that capture the user's health signs and fall detection sensors are also used. The fall detection sensor is an accelerometer that is used to determine whether the user has fallen abruptly, such that warnings signals can be generated to provide the needed car.

Specifications –

1. Arduino UNO- 8 bit ATmega328P (Freq-16MHZ)
2. Temperature Sensor – LM35
3. Pulse Sensor(SEN-11574) - 0.625" Diameter and 0.125" Thick
4. Accelerometer- MMA8452Q Triple Axis , 6 pin Accelerometer
5. Emergency button- 6x6x5mm Tactile Push Button
6. 16x2 LCD Display
7. WiFi Module ESP8266
8. Android/IOS Mobile
9. Power Bank/9V battery
10. Buzzer

Software Requirements –

- Arduino Software IDE
- To Design the ARM7 Hardware. Make circuit in proteus, PCB making, Soldering, H/W testing, Software coding on Embedded C using KEIL Compiler.
- To design an IOT SERVER software in Fire Fox. To design a GUI containing Tables and Graphs.
- MQTT for IOT based applications.

Applications –

- Monitor the motions and status of the person, heart beat rate, temperature, up- to-date.
- A Sudden fall due to heart attacks or loosing balance, or abnormal fall or rise in heart beat rate it alters the nearby ones of the person to get immediate attention
- The Health band also measures the temperature of the person in order to understand the person health condition

